

```
# Import required libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Display plots inside the notebook
%matplotlib inline
```

```
df = pd.read_csv('Mall_Customers.csv')
```

```
df.head()
```

	CustomerID	Gender	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

```
df.shape
```

```
(200, 5)
```

```
df.columns
```

```
Index(['CustomerID', 'Gender', 'Age', 'Annual Income (k$)',  
      'Spending Score (1-100)'],  
      dtype='object')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 200 entries, 0 to 199  
Data columns (total 5 columns):  
#   Column                                Non-Null Count  Dtype  
---  ---                                -  
0   CustomerID                           200 non-null    int64  
1   Gender                               200 non-null    object  
2   Age                                   200 non-null    int64  
3   Annual Income (k$)                   200 non-null    int64  
4   Spending Score (1-100)               200 non-null    int64  
dtypes: int64(4), object(1)  
memory usage: 7.9+ KB
```

```
df.isnull().sum()
```

	0
CustomerID	0
Gender	0
Age	0
Annual Income (k\$)	0
Spending Score (1-100)	0

dtype: int64

```
df.describe()
```

	CustomerID	Age	Annual Income (k\$)	Spending Score (1-100)
count	200.000000	200.000000	200.000000	200.000000
mean	100.500000	38.850000	60.560000	50.200000
std	57.879185	13.969007	26.264721	25.823522
min	1.000000	18.000000	15.000000	1.000000
25%	50.750000	28.750000	41.500000	34.750000
50%	100.500000	36.000000	61.500000	50.000000
75%	150.250000	49.000000	78.000000	73.000000
max	200.000000	70.000000	137.000000	99.000000

```
df.duplicated().sum()
```

```
np.int64(0)
```

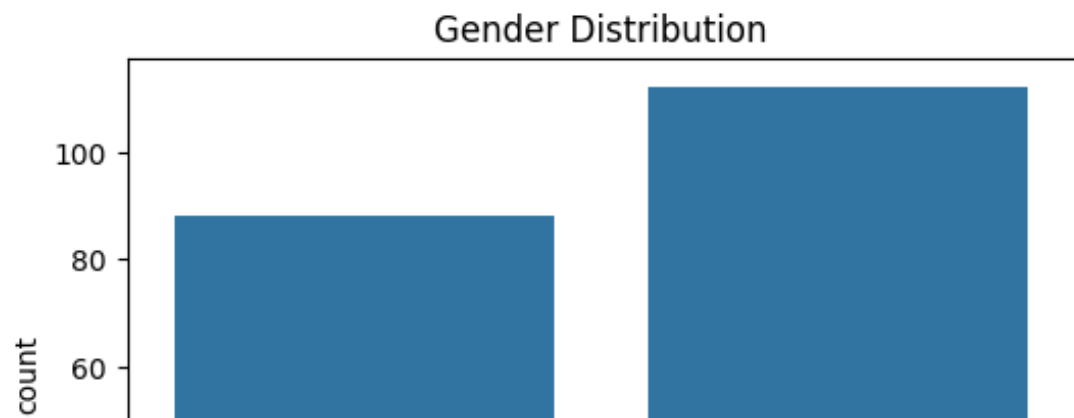
```
df.sample(5)
```

	CustomerID	Gender	Age	Annual Income (k\$)	Spending Score (1-100)
150	151	Male	43	78	17
41	42	Male	24	38	92
78	79	Female	23	54	52
0	1	Male	19	15	39
47	48	Female	27	40	47

```
df.head()
```

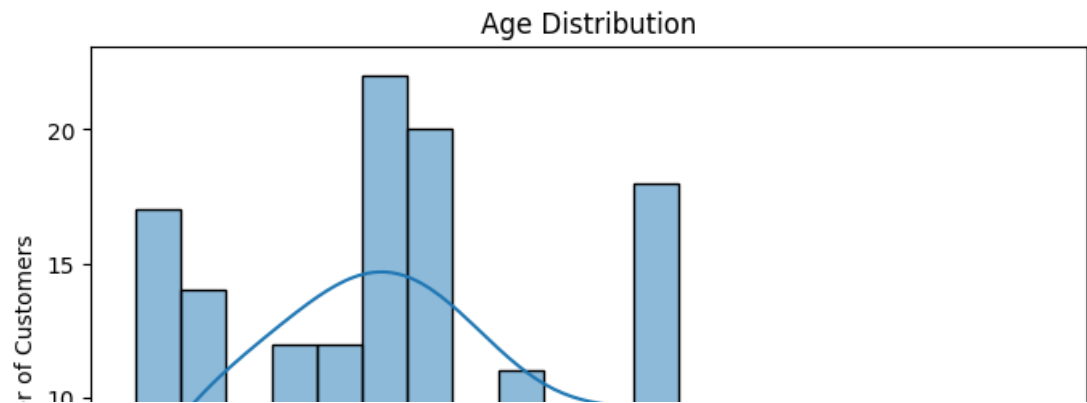
	CustomerID	Gender	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

```
plt.figure(figsize=(6,4))  
sns.countplot(x='Gender', data=df)  
  
plt.title('Gender Distribution')  
plt.show()
```



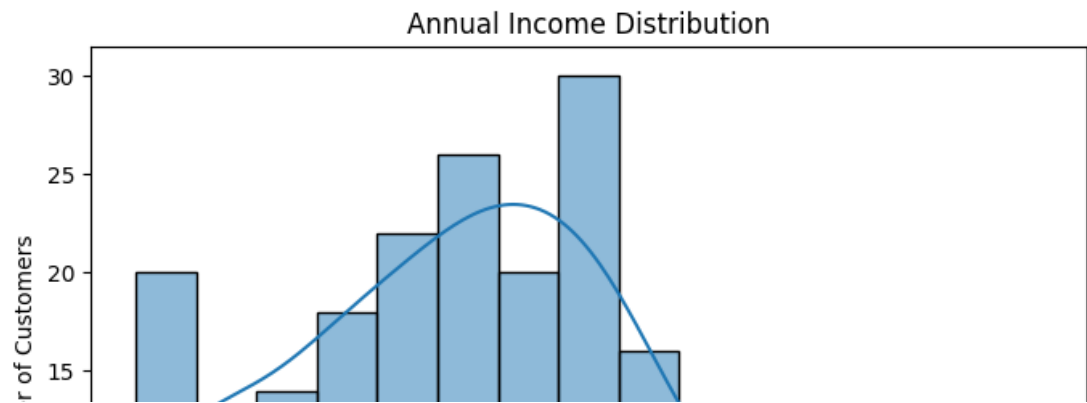
```
plt.figure(figsize=(8,5))
sns.histplot(df['Age'], bins=20, kde=True)

plt.title('Age Distribution')
plt.xlabel('Age')
plt.ylabel('Number of Customers')
plt.show()
```



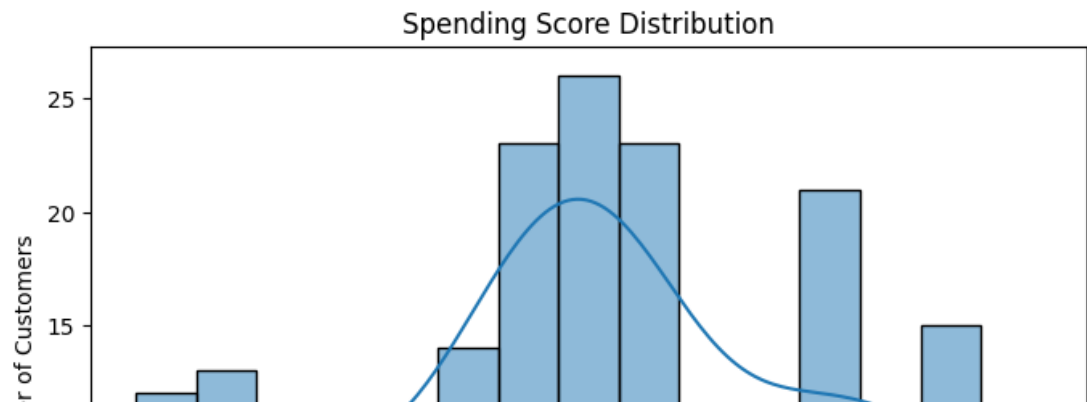
```
plt.figure(figsize=(8,5))
sns.histplot(df['Annual Income (k$)'], bins=15, kde=True)

plt.title('Annual Income Distribution')
plt.xlabel('Annual Income (k$)')
plt.ylabel('Number of Customers')
plt.show()
```



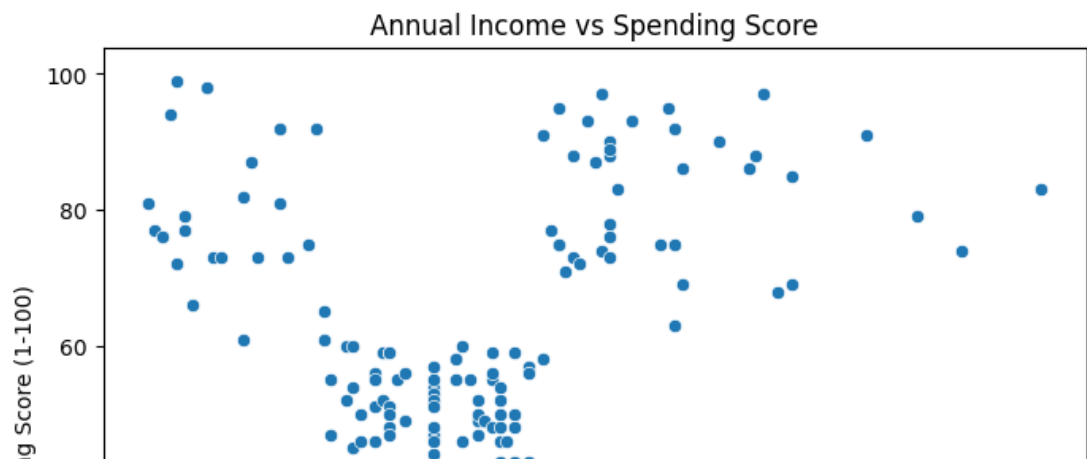
```
plt.figure(figsize=(8,5))
sns.histplot(df['Spending Score (1-100)'], bins=15, kde=True)

plt.title('Spending Score Distribution')
plt.xlabel('Spending Score')
plt.ylabel('Number of Customers')
plt.show()
```



```
plt.figure(figsize=(8,6))
sns.scatterplot(
    x='Annual Income (k$)',
    y='Spending Score (1-100)',
    data=df
)

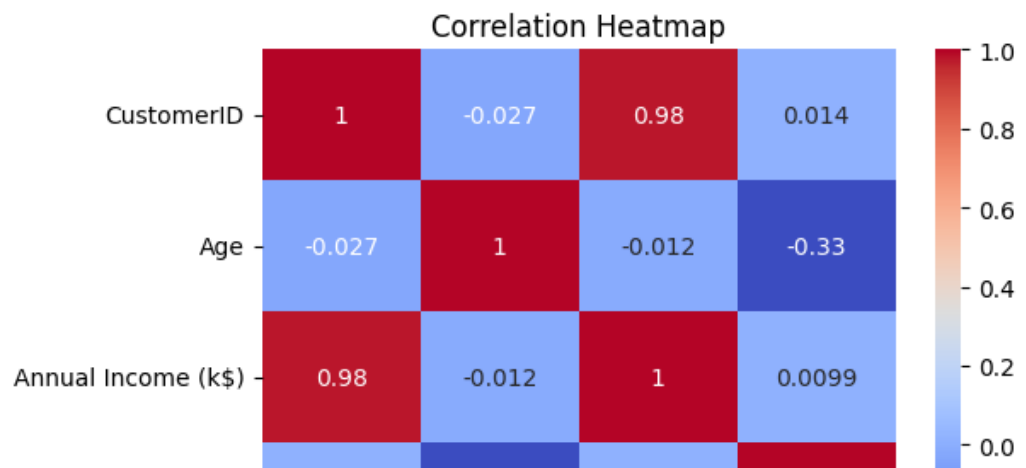
plt.title('Annual Income vs Spending Score')
plt.show()
```



```
plt.figure(figsize=(6,4))

sns.heatmap(
    df.select_dtypes(include=['number']).corr(),
    annot=True,
    cmap='coolwarm'
)

plt.title('Correlation Heatmap')
plt.show()
```



```
X = df[['Annual Income (k$)', 'Spending Score (1-100)']]

# Display the first 5 rows
X.head()
```

	Annual Income (k\$)	Spending Score (1-100)
0	15	39
1	15	81
2	16	6
3	16	77
4	17	40

```
from sklearn.cluster import KMeans
```

```
wcss = []
```

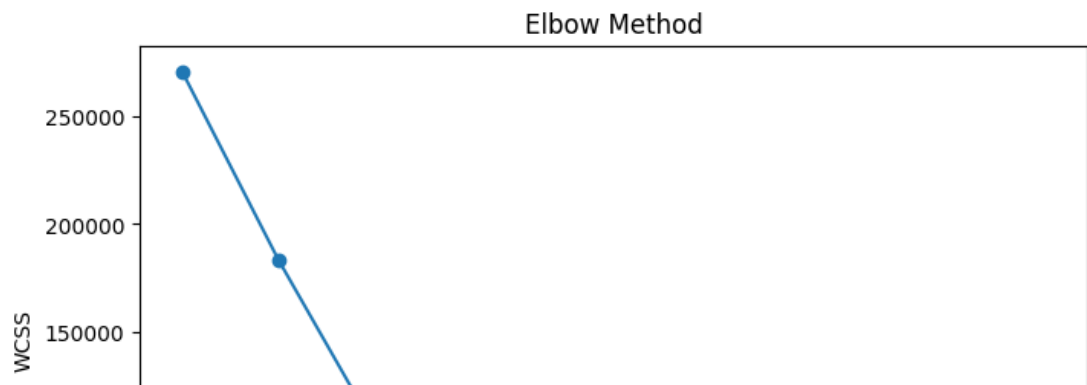
```
for i in range(1, 11):
    kmeans = KMeans(n_clusters=i, random_state=42)
    kmeans.fit(X)
    wcss.append(kmeans.inertia_)
```

```
plt.figure(figsize=(8,5))

plt.plot(range(1,11), wcss, marker='o')

plt.title('Elbow Method')
plt.xlabel('Number of Clusters')
plt.ylabel('WCSS')

plt.show()
```



```
kmeans = KMeans(n_clusters=5, random_state=42)

y_kmeans = kmeans.fit_predict(X)
```

[illegible]

```
df['Cluster'] = y_kmeans
```

```
# Plot the customer segments
plt.figure(figsize=(10, 7))

plt.scatter(X.iloc[y_kmeans == 0, 0], X.iloc[y_kmeans == 0, 1],
            s=100, c='red', label='Cluster 1')

plt.scatter(X.iloc[y_kmeans == 1, 0], X.iloc[y_kmeans == 1, 1],
            s=100, c='blue', label='Cluster 2')

plt.scatter(X.iloc[y_kmeans == 2, 0], X.iloc[y_kmeans == 2, 1],
            s=100, c='green', label='Cluster 3')

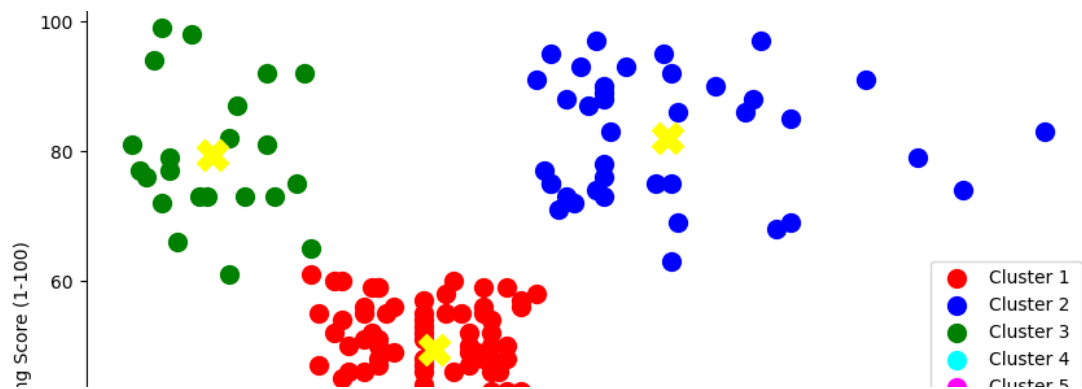
plt.scatter(X.iloc[y_kmeans == 3, 0], X.iloc[y_kmeans == 3, 1],
            s=100, c='cyan', label='Cluster 4')

plt.scatter(X.iloc[y_kmeans == 4, 0], X.iloc[y_kmeans == 4, 1],
            s=100, c='magenta', label='Cluster 5')

# Plot the centroids
plt.scatter(kmeans.cluster_centers_[:, 0],
            kmeans.cluster_centers_[:, 1],
            s=300,
            c='yellow',
            label='Centroids',
            marker='X')

plt.title('Customer Segments')
plt.xlabel('Annual Income (k$)')
plt.ylabel('Spending Score (1-100)')
plt.legend()

plt.show()
```



```
print(kmeans.cluster_centers_)
```

```
[[55.2962963  49.51851852]  
 [86.53846154 82.12820513]  
 [25.72727273 79.36363636]  
 [88.2        17.11428571]  
 [26.30434783 20.91304348]]
```

```
df['Cluster'].value_counts().sort_index()
```

count	
Cluster	
0	81
1	39
2	22
3	35
4	23

dtype: int64

```
df.groupby('Cluster')[['Annual Income (k$)', 'Spending Score
```

	Annual Income (k\$)	Spending Score (1- 100)	Age
Cluster			
0	55.296296	49.518519	42.716049
1	86.538462	82.128205	32.692308
2	25.727273	79.363636	25.272727
3	88.200000	17.114286	41.114286
4	26.304348	20.913043	45.217391

```
df.head(10)
```

	CustomerID	Gender	Age	Annual Income (k\$)	Spending Score (1- 100)	Cluster
0	1	Male	19	15	39	4
1	2	Male	21	15	81	2
2	3	Female	20	16	6	4
3	4	Female	23	16	77	2
4	5	Female	31	17	40	4
5	6	Female	22	17	76	2
6	7	Female	35	18	6	4
7	8	Female	23	18	94	2
8	9	Male	64	19	3	4
9	10	Female	30	19	72	2

```
df.to_csv('Mall_Customers_Clustered.csv', index=False)
```

```
print("Clustered dataset saved successfully!")
```

```
Clustered dataset saved successfully!
```

